

Validity Analysis: NLLB

Target context: Amhara Region Agricultural and Microfinance Translation Cohort

Overall risk: **HIGH**

Deployment Context

Use case: Document-level translation of agricultural input subsidy notices, land-use policy updates, and credit-scheme terms from federal and regional bureaus into Amharic

Target population: Farmer cooperative leaders and rural microfinance clients (e.g., ACSI, ADCSI borrowers) in Amhara interpreting government and lender document

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology $\triangle$	2	Significant gaps	high
Input Content $\triangle$	2	Significant gaps	high
Input Form $\checkmark$	4	Minor gaps	high
Output Ontology $\triangle$	2	Significant gaps	high
Output Content $\triangle$	2	Significant gaps	high
Output Form $\checkmark$	4	Minor gaps	high
Average	2.7		

$\triangle$ = highest concern $\checkmark$ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

Flores-200/NLLB provides credible language- and script-level support for Amharic — Ge'ez script is included with a dedicated SentencePiece vocabulary [Q382], and the benchmark's text-only output form aligns with the deployment's accepted format [elicitation A4]. These are real strengths for this text-only Amharic deployment. However, the four HIGH-priority dimensions (Input Ontology, Input Content, Output Ontology, Output Content) show substantial validity gaps. The benchmark's task taxonomy excludes agricultural, land-tenure, and microfinance genres entirely [Q95, Q166-Q169]; its content is exclusively Wikimedia-derived [Q95, Q151] with the authors themselves acknowledging cultural-coverage limitations [Q151]; its output ontology offers only general translation-quality scalars [Q455, Q648] that cannot detect the high-consequence error types

(numerical-threshold alteration, conditional-logic distortion, terminology drift) that matter for binding administrative documents; and its annotator workforce is professional generalist translators [Q97-Q103], not the federal bureau translators with Ethiopian government-domain expertise that the deployer identifies as the validation authority [elicitation A3]. Independent 2025 research [WEB-10] empirically confirms that Flores+ scores fail to predict out-of-domain translation quality and that chrF++/BLEU can reward superficial named-entity copying. Web search found no Ethiopian agricultural/administrative MT benchmark exists [WEB-3, WEB-4] and no public ACSI/ADCSI Amharic terminology resource [WEB-15, WEB-16], meaning the gaps are real and unaddressed by the broader research community. Aggregate NLLB scores on Flores-200 should be treated as a coarse general-domain capability signal for Amharic, not as evidence of fitness for binding subsidy, land-policy, or credit documents.

ID	Evidence
Q95	“As a significant extension of Flores-101, Flores-200 consists of 3001 sentences sampled from English-language Wikimedia projects for 204 total languages. Approximately one third of sentences are collected from each of these sources: Wikinews, Wikijunior, and Wikivoyage. The content is professionally translated into 200+ languages to create Flores-200. As we translate the same set of content into all languages, Flores-200 is a many-to-many multilingual benchmark.” (p.19)
Q97	“Translators are required to have at least two to three years of translation experience in the relevant language pair if they have an academic degree in translation or linguistics and three to five years of translation experience if they do not have any relevant academic qualification.” (p.20)
Q98	“Translators also undergo a translation test every 18 months to assess their translation quality.” (p.20)
Q99	“Flores-200 reviewers are also required to be native speakers of the target language.” (p.20)
Q100	“Reviewers typically have a translation degree, at least five years of experience working as a translator, translation review experience, and where possible are accredited by a relevant translation board.” (p.20)
Q101	“We note that these are stringent standards, and extensions of Flores-200 to even more low-resource languages in the future may be difficult.” (p.20)
Q102	“Already for many languages, finding a reviewer that meets the criteria above is very challenging.” (p.20)
Q103	“In these cases, we modified the qualification process to accept applications from reviewers with more general language degrees such as Linguistics or African Language Studies, or no degree provided they have had extensive commercial translation experience (e.g. >10 years).” (p.20)
Q151	“We note that like Flores-200, NLLB-Seed's source data is English-centric and sampled from English Wikipedia. This has an important effect: the content reflects what Wikipedia editors find is relevant for English Wikipedia, and likely does not cover diverse content from different cultures.” (p.24)
Q166	“We translate the English side of the WMT21 English-German development set, containing a sample of newspapers from 2020 (Akhbardeh et al., 2021).” (p.25)
Q167	“We translate text extracted from a series of scripted English-language talks covering a variety of topics.” (p.25)
Q168	“We extract 3,000 utterances from the multi-session chat dataset of Xu et al. (2022), which contains on average 23 words per turn.” (p.25)
Q169	“We translated one World Health Organisation report (Donaldson and Rutter, 2017) and combined it with sentences translated from the English portion of the TAUS Corona Crisis Report.” (p.25)
Q382	“We then trained an encoder for Amharic and Tigrinya only, paired with English as in all our experiments, and a specific 8k SentencePiece vocabulary to better support the Ge'ez script.” (p.45)
Q455	“We use the chrF++ metric to compare the model performance (see Section 7.1).” (p.53)
Q648	“In this work, we primarily evaluate using chrF++ using the settings from sacrebleu. However, when comparing to other published works, we utilize BLEU and spBLEU where appropriate.” (p.73)
WEB-3	No Ethiopian agricultural or administrative-domain Amharic benchmark exists (https://arxiv.org/abs/2403.13737)
WEB-4	EthioMT parallel corpora (15 Ethiopian languages) drawn from religious and general domains, not administrative/financial (https://arxiv.org/html/2403.19365v1)
WEB-10	2025 study: mock translations merely reproducing named entities achieved non-trivial BLEU and ChrF++ scores, confirming metrics can reward superficial overlap rather than translational adequacy (https://arxiv.org/html/2508.20511v1)
WEB-15	ACSI official website — no Amharic financial terminology guide publicly available (http://www.acsi.org.et/)
WEB-16	FinDev Gateway ACSI profile — no terminology resource (https://www.findevgateway.org/case-study/2004/01/amhara-credit-and-saving-institution-acsi-institutional-profile-current-status)

Practical Guidance

What This Benchmark Measures

NLLB/Flores-200 measures general-domain sentence-level translation quality for Amharic prose, anchored in English-Wikimedia-derived content [Q95] and scored with chrF++/spBLEU/calibrated XSTS [Q455, Q648, Q654]. The strongest dimensions for this deployment are Input Form and Output Form: the benchmark validly assesses whether a model can produce coherent Amharic text in Ge'ez script for general-purpose prose, which is a necessary precondition for the deployment.

Construct Depth

The benchmark probes general translation capability shallowly across many directions rather than deeply for any one domain. For Amharic specifically, evaluation is limited to 3,001 Wikimedia-derived sentences [Q95] scored with sentence-level metrics [Q336] that 2025 evidence shows can reward superficial overlap [WEB-10]. There is no evidence depth on: (a) terminology consistency across a multi-clause document, (b) numerical-threshold fidelity, (c) conditional-logic preservation, (d) acceptance by Ethiopian federal bureau translators, or (e) any agricultural/financial/bureaucratic register. The benchmark's evidence stops well short of what is needed for binding administrative document deployment.

What Else You Need

To achieve a complete assessment for this deployment, the user should commission: (1) a domain-specific Amharic evaluation set covering agricultural input subsidies, land-tenure proclamations, and ACSI/ADCSI credit documents (Input Ontology + Input Content gaps); (2) a domain rubric with rules for numerical-fidelity, terminology-consistency, and conditional-logic preservation, since chrF++ cannot detect these error types (Output Ontology gap); (3) a federal-bureau-translator annotator pool to produce reference translations and acceptability ratings for representative documents (Output Content gap); (4) document-level coherence and terminology-consistency scoring on multi-clause samples (Output Form residual gap); and (5) a targeted Amharic Flores-200 audit analogous to those done for Hausa/Sepedi/Xitsonga/isiZulu [WEB-14], since no such audit has been published.

ID	Evidence
Q95	"As a significant extension of Flores-101, Flores-200 consists of 3001 sentences sampled from English-language Wikimedia projects for 204 total languages. Approximately one third of sentences are collected from each of these sources: Wikinews, Wikijunior, and Wikivoyage. The content is professionally translated into 200+ languages to create Flores-200. As we translate the same set of content into all languages, Flores-200 is a many-to-many multilingual benchmark." (p.19)
Q336	"We look for best performance on the Flores-200 development set, and report detokenized BLEU on Flores-200 devtest." (p.41)
Q455	"We use the chrF++ metric to compare the model performance (see Section 7.1)." (p.53)
Q648	"In this work, we primarily evaluate using chrF++ using the settings from sacrebleu. However, when comparing to other published works, we utilize BLEU and spBLEU where appropriate." (p.73)
Q654	"XSTS rates each source sentence and its machine translation on a five-point scale, where 1 is the lowest score and 5 is the highest score." (p.74)
WEB-10	2025 study: mock translations merely reproducing named entities achieved non-trivial BLEU and ChrF++ scores, confirming metrics can reward superficial overlap rather than translational adequacy (https://arxiv.org/html/2508.20511v1)
WEB-14	Post-publication FLORES-200 corrections exist for Hausa, Sepedi, Xitsonga, isiZulu — but not Amharic, leaving Amharic reference unaudited (https://www.emergentmind.com/topics/flores-200-benchmark-dataset)